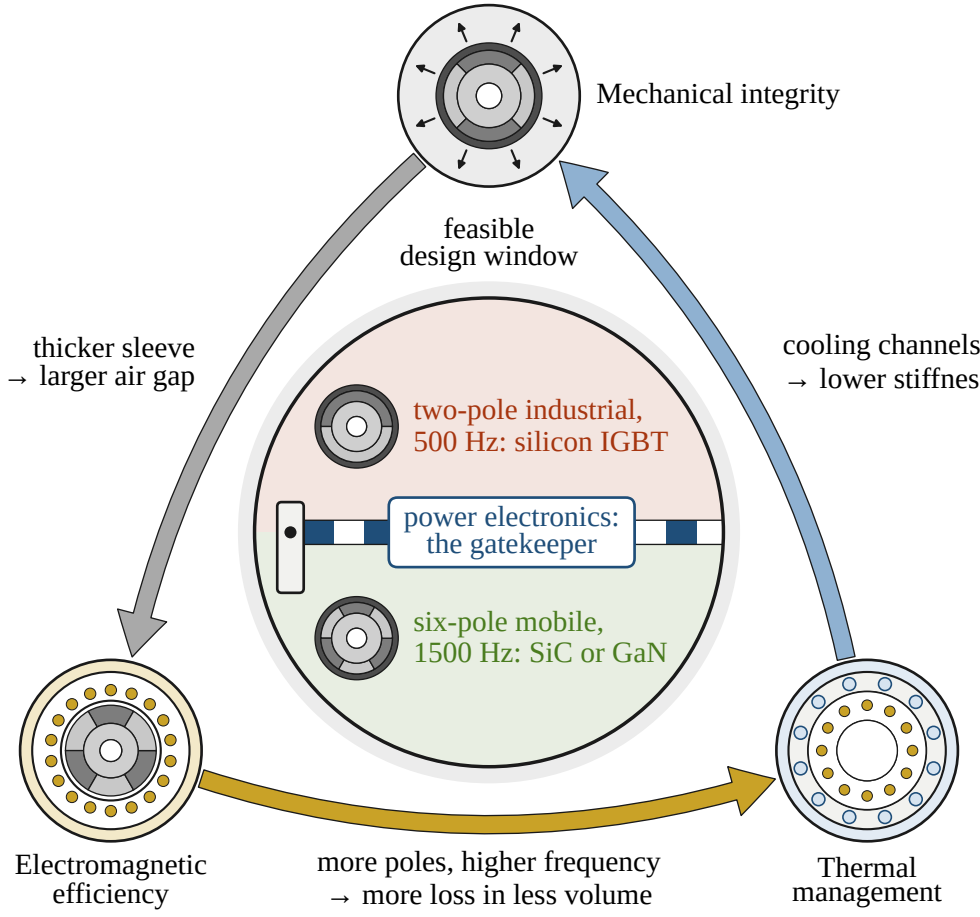
Mechanical integrity
feasible
design window
thicker sleeve
→ larger air gap
cooling channels
→ lower stiffness
two-pole industrial,
500 Hz: silicon IGBT
power electronics:
the gatekeeper
six-pole mobile,
1500 Hz: SiC or GaN
Electromagnetic
efficiency
more poles, higher frequency
→ more loss in less volume
Thermal
management